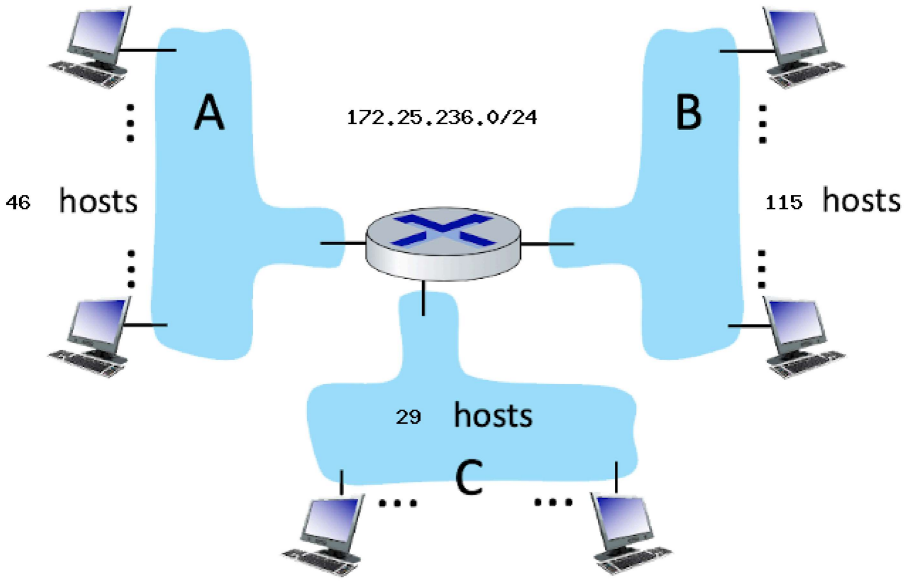
INTERACTIVE END-OF-CHAPTER EXERCISES

SUBNET ADDRESSING

Consider the router and the three attached subnets below (A, B, and C). The number of hosts is also shown below. The subnets share the 24 high-order bits of the address space: 172.25.236.0/24



Assign subnet addresses to each of the subnets (A, B, and C) so that the amount of address space assigned is minimal, and at the same time leaving the largest possible contiguous address space available for assignment if a new subnet were to be added. Then answer the questions below.

QUESTION LIST

1. Is the address space public or private?
2. How many hosts can there be in this address space?
3. What is the subnet address of subnet A? (CIDR notation)
4. What is the broadcast address of subnet A?
5. What is the starting address of subnet A?
6. What is the ending address of subnet A?
7. What is the subnet address of subnet B? (CIDR notation)
8. What is the broadcast address of subnet B?
9. What is the starting address of subnet B?
10. What is the ending address of subnet B?
11. What is the subnet address of subnet C? (CIDR notation)
12. What is the broadcast address of subnet C?
13. What is the starting address of subnet C?
14. What is the ending address of subnet C?

SOLUTION

1. The address 172.25.236.0/24 is private.

2. Maximum number of hosts = 2^x - 2 = 2^8 - 2 = 254. The reason we have to subtract 2 from the final number is because there are always 2 addresses allocated for each address block: the subnet ID (the first address) and the broadcast address (the last address); for example, if you have 5 bits for hosts, you can have 30 hosts, because 2 of the addresses are for the subnet ID and the broadcast address which when added equals 32, which is 2^5.

3. Subnet A has 46 hosts, so it will need at least 48 addresses (for the subnet ID and broadcast address). The least number of bits that satisfy this is 6 bits. Knowing that, we take the prior subnet and add 64, the result of which is 172.25.236.128/26

4. The broadcast address of subnet A (172.25.236.128/26) is 172.25.236.191, because it is the last address in the IP range.

5. The first IP address of subnet A (172.25.236.128/26) is 172.25.236.129, found by adding 1 to the subnet address.

6. The last IP address of subnet A (172.25.236.128/26) is 172.25.236.190, found by subtracting 1 from the broadcast address (172.25.236.191).

7. Similar to the prior subnet, subnet B has 115 hosts, so it will need at least 117 addresses (for the subnet ID and broadcast address). The least number of bits that satisfy this is 7 bits. Knowing that, we take the prior subnet and add 128, the result of which is 172.25.236.0/25

8. The broadcast address of subnet B (172.25.236.0/25) is 172.25.236.127, because it is the last address in the IP range.

9. The first IP address of subnet B (172.25.236.0/25) is 172.25.236.1, found by adding 1 to the subnet address.

10. The last IP address of subnet B (172.25.236.0/25) is 172.25.236.126, found by subtracting 1 from the broadcast address (172.25.236.127).

11. Similar to the prior two subnets, subnet C has 29 hosts, so it will need at least 31 addresses (for the subnet ID and broadcast address). The least number of bits that satisfy this is 5 bits. Knowing that, we take the prior subnet and add 32, the result of which is 172.25.236.192/27

12. The broadcast address of subnet C (172.25.236.192/27) is 172.25.236.223, because it is the last address in the IP range.

13. The first IP address of subnet C (172.25.236.192/27) is 172.25.236.193, found by adding 1 to the subnet address.

14. The last IP address of subnet C (172.25.236.192/27) is 172.25.236.222, found by subtracting 1 from the broadcast address (172.25.236.223).

QUESTION 1 OF 14

Is the address space public or private?

Answer

CHECK HINT SKIP

We gratefully acknowledge the programming and problem design work of John Broderick (UMass '21), which has really helped to substantially improve this site.

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Comments welcome and appreciated: kurose@cs.umass.edu